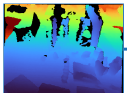


## Inputs

intrinsics  $\mathbf{K}$



keyframe  $\mathcal{I}^k$  and  
frame  $\mathcal{I}^f$



sensor depth  $\mathbf{Z}$

FRONT END every frame

1

### Depth-conditioned Pow3R

two-view pointmaps and confidences  
dense even where  
the sensor has holes

*cond*

2

### Metric scale

from the  
sensor depth

*scale*

3

### Tracking and keyframing

Sim(3) pose from projective matches  
new keyframe when overlap drops

new keyframe

lost

relocalized

5

### Depth-anchored backend

keyframe geometry held  
to the sensor depth  
all keyframe poses refined together

*anchor*

4

### Loop closure, relocalization

retrieval on Pow3R's tokens  
verified by projective matching

edges

BACKEND every new keyframe

final poses

## Outputs

metric trajectory



estimate vs. ground truth

dense metric map



keyframes + map-only keyframes